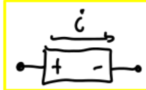
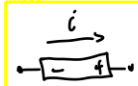
Rappel

$$q = 1.6022 \times 10^{-19}\ C$$





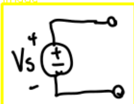
$$v = \frac{dw}{dq}$$

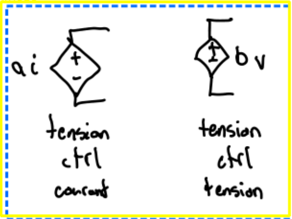
$$i = \frac{dq}{dt}$$

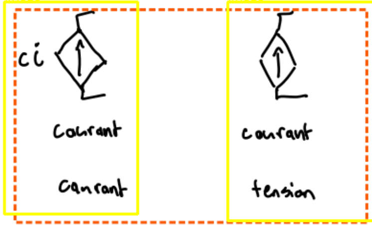
$$p = +i\ v$$

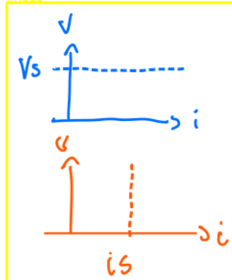
ou

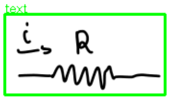
$$-i\ v$$







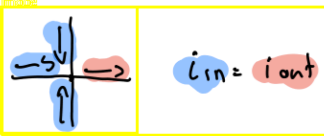


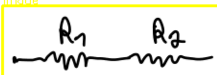


$V = Ri$

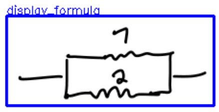
$$p = Ri^2 = \frac{v^2}{R}$$

$G = 1/R$

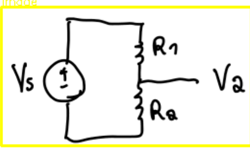




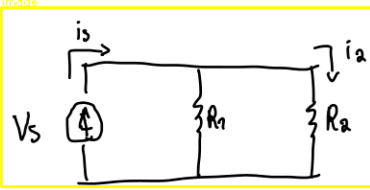
$$R = R_1 + R_2$$



$$R = \left(1/R_1 + 1/R_2\right)^{-1}$$



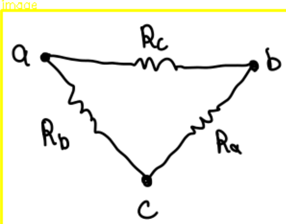
$$V_2 = V_s \cdot \frac{R_2}{R_1 + R_2}$$

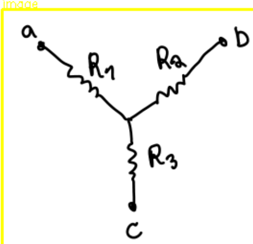


$$i_2 = i_s \cdot \frac{R_1}{R_1 + R_2}$$

Trans phorm

$$\Delta \quad Y$$





$$R_{ab} = \frac{R_c(R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{ac} = \frac{R_b(R_a + R_c)}{R_a + R_b + R_c} = R_1 + R_3$$

$$R_{bc} = \frac{R_a(R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$\Delta \to Y$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$Y \to \Delta$$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_3}$$

Vocabulaire :

Noeud : point où 2+ composants se rejoignent

Noeud essentiel : 3+ compo

parcours : chemin avec no compo en double

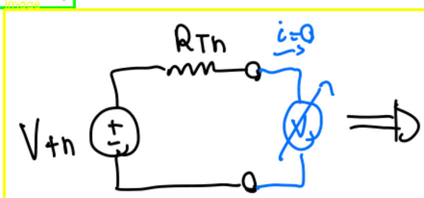
Branche : parcours reliant 2 noeuds

Branche ess : parcours relie 2 noeuds essentiels sans passer par autre ess

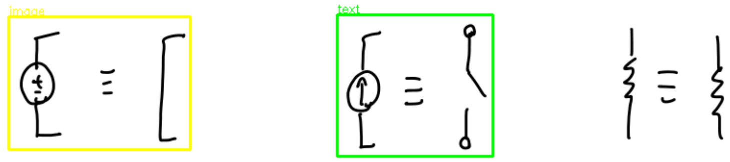
Boucle : parcours fermé

Maille : boucle sans autres boucle

Thev

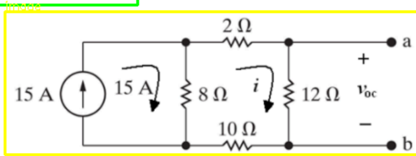


pour trouver rth, mod circuit avec :



Calculer l'équivalent de Thévenin – Exemple (1)

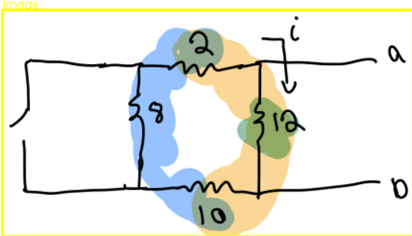
Tension en circuit ouvert



$2i + 12i + 10i + 8(i - 15) = 0$

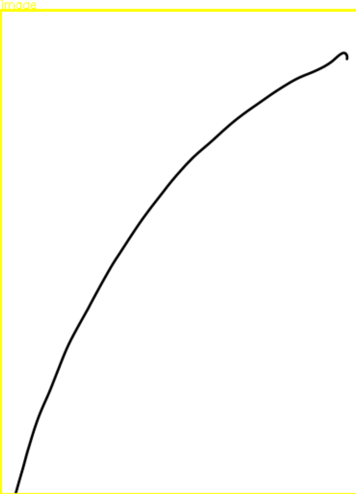
$\therefore i(2 + 12 + 10 + 8) = (8)(15) \quad \Rightarrow i = 3.75\ A$

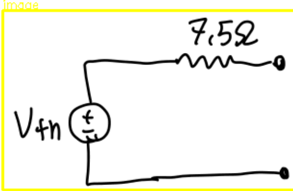
$$v_{oc} = 12i = 45\ \ V = V_{Th}$$



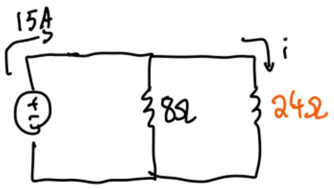
R = 20Ω

$$\left(\frac{1}{12} + \frac{1}{20}\right)^{-1} = 7{,}5 = R_{th}$$



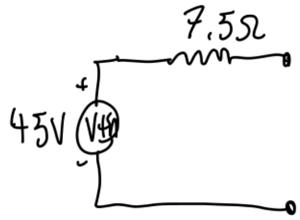


. Trouver V_{th}



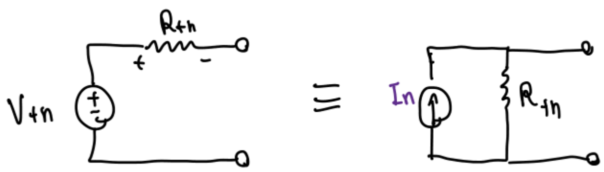
$i = 15 \times \frac{8}{24+8} = 3,75$ Amp

↳ $V_A = 3,75 \times 12 = 45V$



← circuit ther equivalent

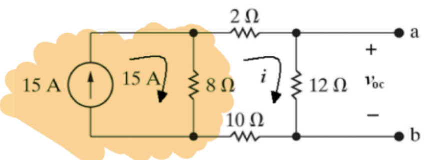
Northon



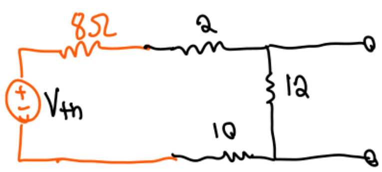
$V_{th} = I_n \times R$

$V_{th} = R_{th} \cdot I_n$

Same exo with Northon

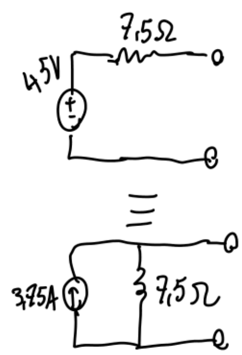


⟶



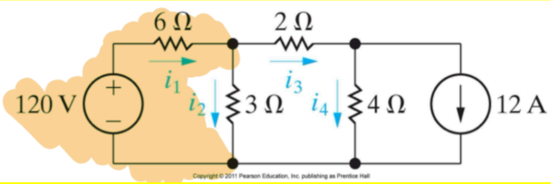
$V_{th} = 8\Omega \cdot 15 = 120V$

ther ⟶



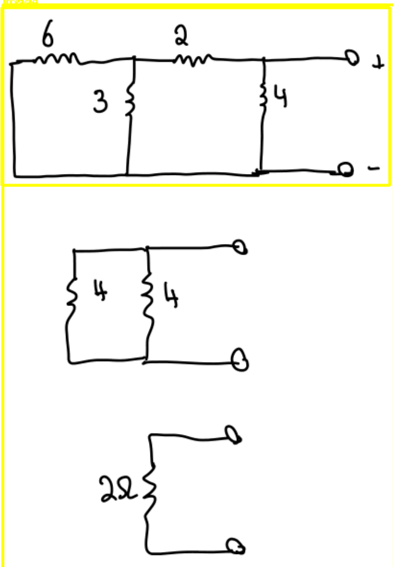
Principe de superposition – Exemple (1)

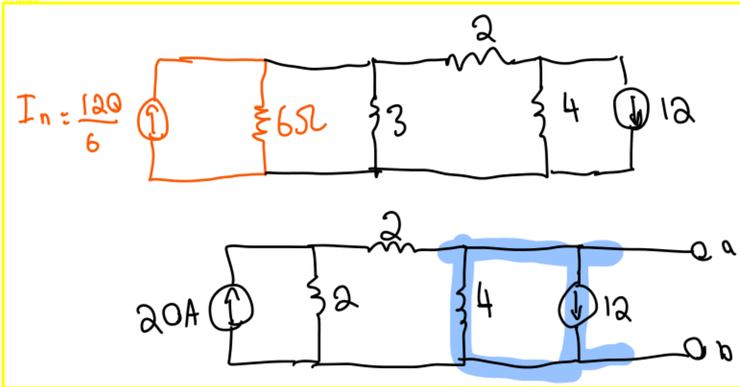
- Calculer le courant i_3

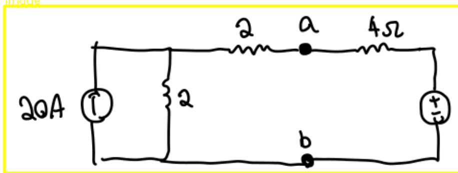


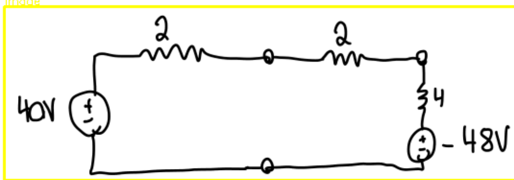
$i_1 = i_3 + i_2$

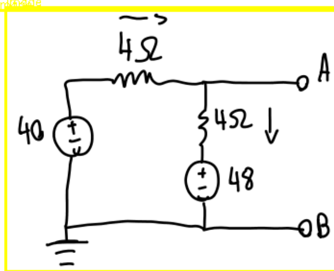
$6 i_1 =$











~~$4i_1 - 4i_2 = 8$~~

~~$i_1 - i_2 = 2$~~

~~$i_2 = -(2 - i_1)$~~

Vocabulaire :

Noeud : point où 2+ composantes se rejoignent

Noeud essentiel : 3+ compo

parcours : chemin avec no compo en double

Branche : parcours reliant 2 noeuds

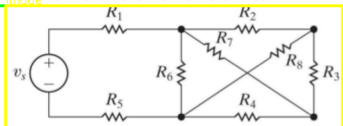
Branche ess : parcours relie 2 noeuds essentiels sans passer par autre ess

Boucle : parcours fermé

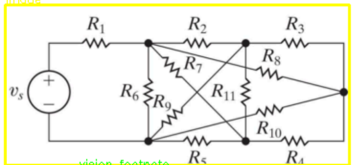
Maille : boucle sans autres boucle

Terminologie

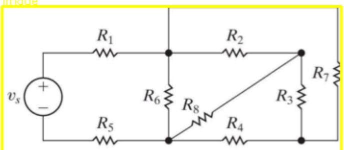
- Circuit planaire: Circuit qui peut se dessiner sur un plan sans que des branches se croisent.



(a)

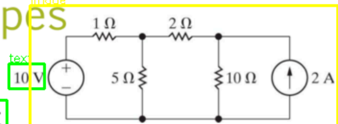


Circuit non-planaire



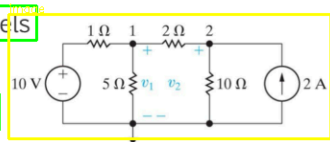
(b)

Méthode des nœuds - Étapes



n_e=3, b_e=5

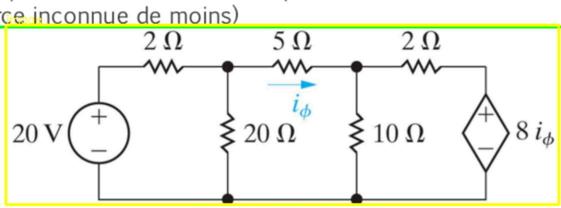
- Basée sur le LKC aux nœuds essentiels
- Circuits planaires ou pas: Redessiner clairement
- Sélectionner un des nœuds essentiels pour en faire la référence
- Nommer (variables) les tensions des nœuds essentiels restants
- Écrire les équations LKC des nœuds essentiels **restants** (n_e-1 équations LKC)
  - Courants inconnus exprimés en fonction des tensions et résistances (éventuellement impédances)
- Solutionner pour les tensions de nœuds essentiels
- Vérifier votre solution
- Déduire d'autres quantités (courants) au besoin



Méthode specific à intra
maybe final too

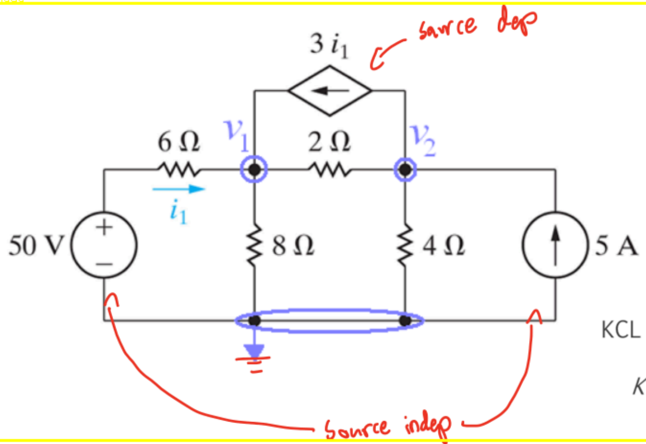
Cas spécial: Sources dépendantes

- La même méthode s'applique s'il y a des sources dépendantes
- Il faut une **équation de substitution** pour chaque source dépendante et qui définit la variable dont dépend la source dépendante **en termes d'autres variables existantes**
  - Cela permet de relier les deux quantités entre elles (donc une variable de source inconnue de moins)



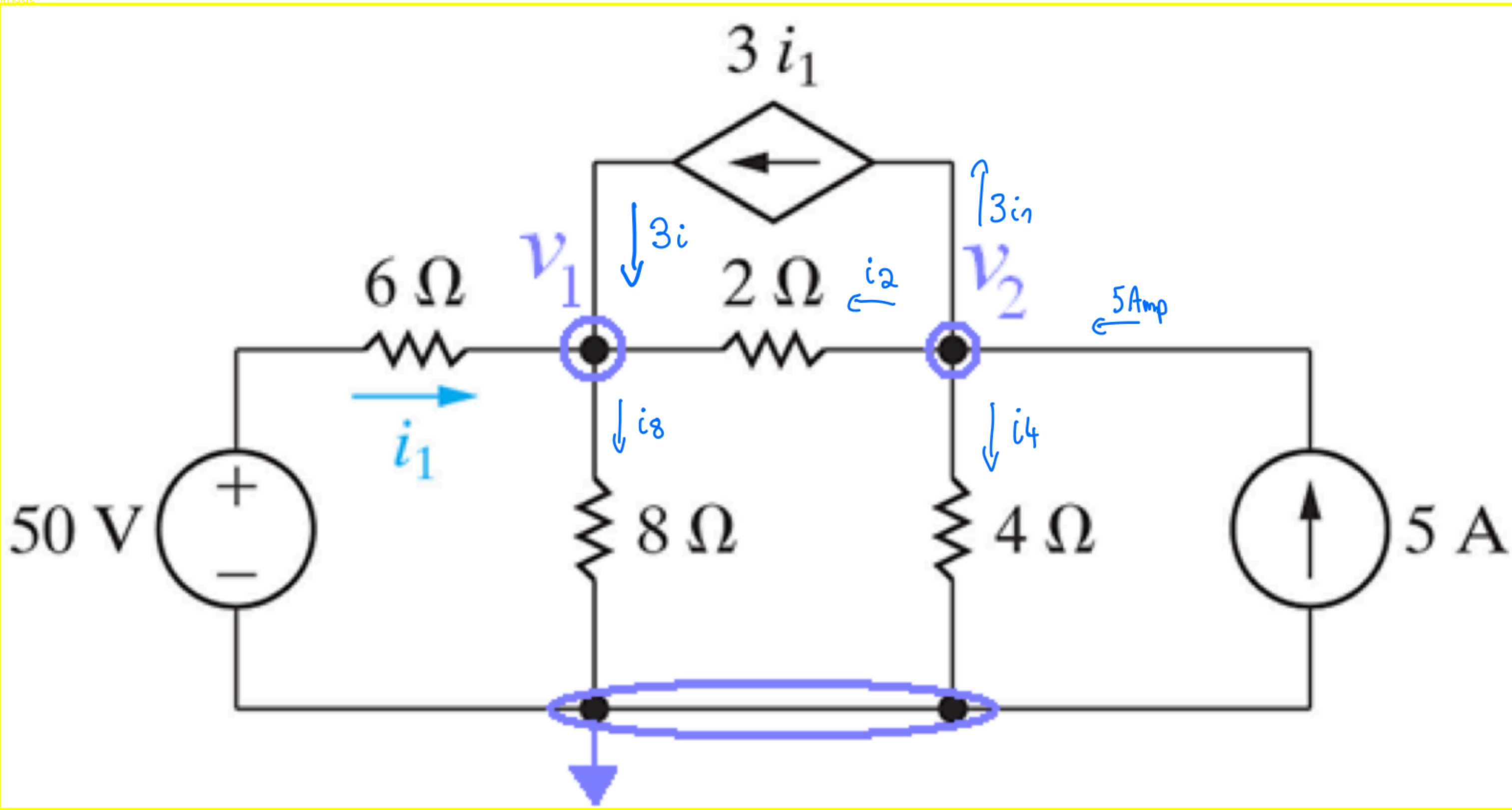
Méthode des nœuds – Exemple 3 (1)

- Trouvez la puissance associée à chaque source.



$$\text{KCL à } v_1\text{:}\quad \frac{v_1-50}{6}+\frac{v_1}{8}+\frac{v_1-v_2}{2}-3i_1=0$$

$$KCL \text{ à } v_2\text{:}\quad \frac{v_2-v_1}{2}+\frac{v_2}{4}-5+3i_1=0$$



V_1: $4i_1 + i_2 = i_8$

V_2: $5 = 3i_1 + i_2 + i_4$

$$i_1 = \frac{50-V_1}{6} \qquad i_2 = \frac{V_2-V_1}{2}$$

$$i_8 = V_1/8 \qquad i_4 = V_2/4$$

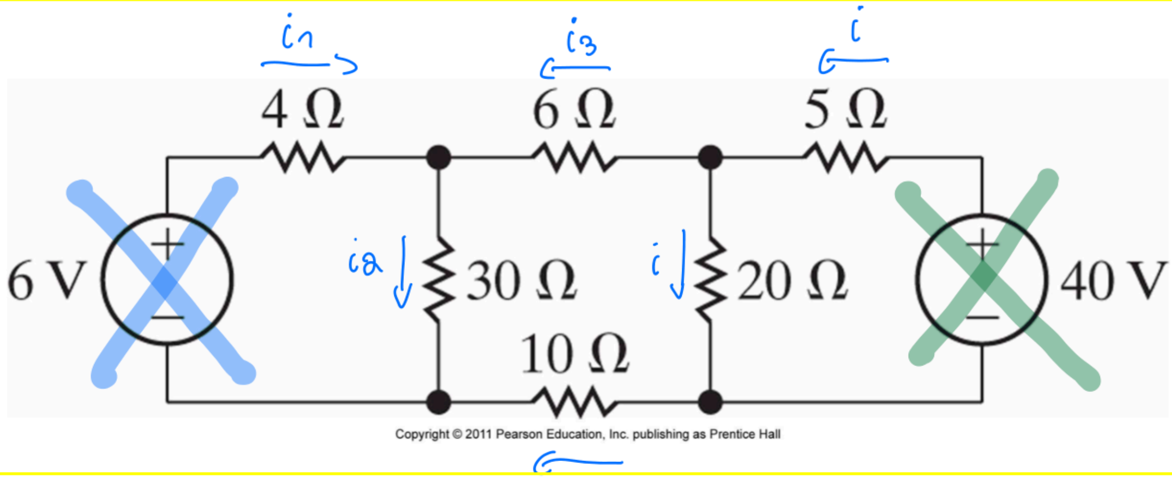
$$\left.\begin{array}{l} V_1\text{:}\quad \frac{4}{6}(50-V_1) + \frac{V_2-V_1}{2} = \frac{V_1}{8} \\ V_2\text{:}\quad 5 = \frac{3}{6}(50-V_1) + \frac{V_2-V_1}{2} + \frac{V_2}{4} \end{array}\right\} \to \text{Ti solve} \Rightarrow \begin{array}{l} V_1 = 32V \\ V_2 = 16V \end{array}$$

$$\left[i_1 = \frac{50-32}{6} = 3A \qquad i_2 = \frac{32-16}{2} = 8A \qquad i_4 = 8A \qquad i_8 = 4A \right.$$

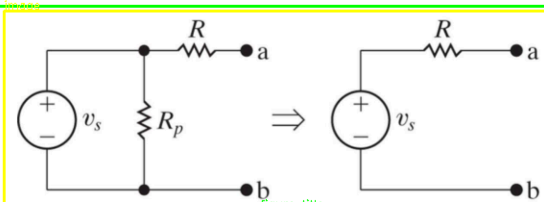
$$p = Vi$$

Simplifier un circuit

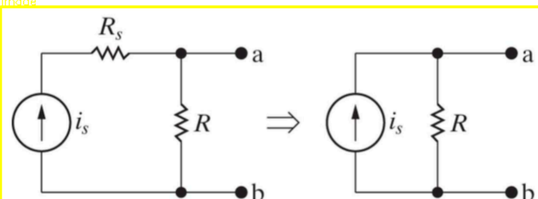
- Est-ce que la source de 6 volts absorbe ou délivre de la puissance? $i = i$ sans bleu $+ i$ sans vert



Cas spéciaux: on veut pouvoir faire les transformations suivantes



(a)



(b)